

GONZALO VOLANTE

SOFTWARE SYSTEMS & APPLIED AI ENGINEER | SOLUTIONS ARCHITECT

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PERFIL PROFESIONAL

Ingeniero de Sistemas de Software e IA Aplicada especializado en el diseño de plataformas de misión crítica, pipelines de ingesta de datos y arquitecturas de automatización de procesos. Trayectoria comprobada en producción diseñando ERPs para el sector salud y plataformas industriales de gestión de energía renovable, manejando más de 1.4 millones de registros clínicos y telemetría de hardware IoT en tiempo real. Sólida experiencia en integración multimodal de LLMs (Google Gemini Vision API), parsing de datos desde formatos semiestructurados (PDFs, facturas de servicios, resúmenes bancarios), mecanismos de caché en memoria y pipelines de despliegue automatizado en entornos Linux.

COMPETENCIAS CLAVE

Arquitectura & Datos:

Domain-Driven Design, SQLite de alta concurrencia (better-sqlite3 sincrónico), memorias intermedias en RAM (vrCache 0 ms de latencia), migración y saneamiento masivo (+1.4M de registros).

Full-Stack & IA Aplicada:

React, TypeScript, Tailwind CSS, Python (Flask), Google Gemini Vision API (OCR multimodal de comprobantes), algoritmos financieros deterministas FIFO, REST APIs & Webhooks.

Infraestructura & Automatización:

Linux VPS (systemd, Nginx, Unicorn, Certbot SSL), Playwright RPA (navegación headless anti-bot), telemetría solar (Growatt API), parsing con pdfplumber, GitHub Actions CI/CD.

EXPERIENCIA PROFESIONAL (SISTEMAS EN PRODUCCIÓN ACTIVA)

Laboratorio Bioquímico SARESA

2024 - Presente · Salta, Argentina

Lead Solutions Architect & Full-Stack Systems Developer | LIS SARESA V4 (Core Laboratory Information System & Medical ERP)

- Architected and deployed LIS SARESA V4, an end-to-end mission-critical laboratory information system and medical ERP serving active healthcare operations.
- Engineered the migration, indexing, and sanitization of +1,400,000 clinical analysis results, +86,000 medical orders, and a registry of +30,000 active patients.
- Implemented an in-memory RAM caching engine (vrCache) that eliminated N+1 database queries, achieving 0 ms latency across 1,000+ age-and-sex-dependent biological reference evaluations.
- Integrated Google Gemini Vision API for automated multimodal extraction of medical reagent supplier invoices and packing slips into validated database records.
- Developed a deterministic FIFO transactional accounting algorithm to automate debt allocation and ledger consolidation across medical supplier accounts.
- Configured automated deployment pipelines on a dedicated Linux VPS utilizing systemd services, synchronous SQLite access, and Nginx with Certbot SSL renewal.

Nuevas Energías

Sep 2024 - Presente · Salta, Argentina

IT Specialist & Software Solutions Developer | Central Operations Platform & Solar Energy Balance Engine

- Designed and put into production the central internal operating platform, replacing fragmented legacy spreadsheets with a unified web-based management system.
- Programmed automated solar photovoltaic sizing algorithms (On-Grid and Off-Grid), reducing proposal turnaround time from 48 hours to under 3 minutes (~90% cycle reduction).
- Built an automated ingestion and document parsing pipeline using Python (pdfplumber) to analyze provincial electricity bills (EDES) and detect contracted power transgressions.
- Integrated solar inverter IoT hardware telemetry (Growatt API) with utility billing data in an isolated SQLite database (plantas.db) for automated net-metering energy balances.
- Built a custom read-only CRM client with local persistent caching to interface with external APIs, eliminating rate-limit penalties while maintaining real-time commercial tracking.
- Implemented full CI/CD deployment automation using GitHub Actions, compiling React/TypeScript frontends and syncing to a production Linux host running Unicorn and Nginx.

PROYECTOS DE INGENIERÍA ESPECIALIZADA

Publi-Prop — Headless Web Automation & Browser Session Orchestrator (Node.js, Playwright, Chromium, Session Storage). Built an automated real estate publishing pipeline capable of bypassing complex web forms through headless Chromium execution and persistent session management.

Auditor EDESA — Electrical Tariff Ingestion & Analytical Engine (Python 3.11, pdfplumber, SQLite, Flask, WeasyPrint). Analytical document parser extracting time-differentiated hourly energy consumption patterns and modeling mathematical penalty curves into structured audit reports.

OTBN Geospatial Platform — Satellite Remote Sensing & GIS (Python, Google Earth Engine API, QGIS, GeoPandas). Spatial analysis scripts supporting the Technical Committee for the Native Forests Law (Ley de Bosques) in Salta, extracting NDVI and NDWI vegetation indices over Sentinel and Landsat imagery.

Notion Statement Importer — Standalone Financial Parsing Engine (Python, Notion API, PyInstaller, REST APIs). Standalone executable ingesting credit card PDF statements with automatic tagging, duplicate filtering, and REST API database synchronization.

INGENIERÍA COLABORATIVA & SOFTWARE FACTORY

YouCreate 2.0 (iGrowker, 2024)

Led frontend engineering squad delivering modular financial balance dashboards using React, TypeScript, and Tailwind CSS.

Frontend Lead & Squad Contributor

Smart Projects (Foo Talent Group, 2024)

Agile team simulation with continuous Git/GitHub branch management, code reviews, and sprint delivery cadences.

Software Developer

FORMACIÓN & CERTIFICACIONES

Certified Tech Developer — Software Engineering Program — Digital House (2021 - 2023)

Diseño UX/UI Especializado — Coderhouse (2021)

IDIOMAS

Español: Nativo | Inglés: B2 Profesional (Capacidad de lectura técnica avanzada, redacción de documentación de arquitectura y comunicación fluida en equipos internacionales).